Solutions to Differential Equations and Fourier Analysis Questions

Here's the complete LaTeX code with answers, options, and solving steps for all questions:

"'latex

1. If 2π is the period of $\sin x$ then $\sin(x+16\pi)=?$
   Solution: Period of $\sin x$ is 2π. $16\pi=8\times2\pi$, so $\sin(x+16\pi)=\sin(x+8\times2\pi)=\sin x$
   Answer: (a) $\sin x$

2. Using variation of parameters, particular integral for $y''+y=csc\,x$
   Solution: $y_c=A\cos x+B\sin x$, Wronskian $W=1$.
   $u_1=-\int\frac{y_2\cdot csc\,x}{W}dx=-\int\sin x\cdot csc\,x\,dx=-x$
   $u_2=\int\frac{y_1\cdot csc\,x}{W}dx=\int\cos x\cdot csc\,x\,dx=\ln|\sin x)$
   $y_p=u_1y_1+u_2y_2=-x\cos x+\sin x\ln|\sin x)$
   Answer: (b) $-x\cos x+\sin x(\log\sin x)$

3. The equation $\frac{dx}{dy}+Px=Q$ is linear if
   Solution: Standard linear form: $\frac{dx}{dy}+P(y)x=Q(y)$. So P,Q must be functions of y only.
   Answer: (a) P and Q are functions of y and constant only

4. Fourier coefficient b_n for $f(x)=x^3$ on $(-\pi,\pi)$
   Solution: $b_n=\frac{2}{\pi}\int_0^{\pi}x^3\sin(nx)dx=2(-1)^n\left[-\frac{\pi^2}{n}+\frac{6}{n^3}\right)$
   Answer: (a) $2(-1)^n\left[-\frac{\pi^2}{n}+\frac{6}{n^3}\right)$

5. Fourier transform of $f(x)=1-x^2$ for $|x|\le1$, else 0
   Solution: $F(s)=\frac{2}{\sqrt{2\pi}}\int_0^1(1-x^2)\cos(sx)dx=\frac{4}{s^3\sqrt{2\pi}}[-s\cos s+\sin s)$
   Answer: (b) $\frac{4}{s^3\sqrt{2\pi}}[-s\cos s+\sin s)$

6. Order and degree of $\frac{d^2y}{dx^2}+\sqrt{1+\left(\frac{dy}{dx}\right)^3}=0$
   Solution: Order = highest derivative = 2. Square both sides: $(y'')^2=1+(y')^3$, degree =

power of highest derivative = 2.
Answer: (d) 2,2

7. Fourier cosine transform of $f(x)=e^{-2x}+4e^{-3x}$
Solution: $F_c(\omega)=\sqrt{\frac{2}{\pi}}\int_0^{\infty}(e^{-2x}+4e^{-3x})\cos(\omega x)\,dx=\sqrt{\frac{2}{\pi}}\left(\frac{2}{4+\omega^2}+\frac{12}{9+\omega^2}\right)$

8. $f(x,y)=\cos x+e^x+y$ is
Solution: Homogeneous function requires $f(tx,ty)=t^n f(x,y)$. Terms $\cos x$ and e^x don't scale with t, so non-homogeneous.
Answer: (b) non-homogeneous function

9. Integrating factor for $\frac{dz}{dy}-4z=2y^2$
Solution: Standard linear: $\frac{dz}{dy}+P(y)z=Q(y)$. Here $P(y)=-4$. IF = $e^{\int P\,dy}=e^{\int -4\,dy}=e^{-4y}$
Answer: $e^{\int -4\,dy}$

10. For $(2x+y)dx=(-x-y)dy$, check condition
Solution: Rewrite: $(2x+y)dx+(x+y)dy=0$. $M=2x+y$, $N=x+y$. $\frac{\partial M}{\partial y}=1$, $\frac{\partial N}{\partial x}=1$. So $\frac{\partial M}{\partial y}=\frac{\partial N}{\partial x}$.
Answer: $\frac{\partial M}{\partial y}=\frac{\partial N}{\partial x}$

11. Value of Wronskian $W(1,x,x^2)$
Solution: $W = \left| \begin{matrix} 1 & x & x^2 \\ 0 & 1 & 2x \\ 0 & 0 & 2 \end{matrix} \right.$ ¿¿$=1\cdot(1\cdot 2-2x\cdot 0)=2$
Answer: (b) 2

12. Particular integral for $y''-y=2x^4-3x+1$
Solution: Try $y_p=Ax^4+Bx^3+Cx^2+Dx+E$. Substituting and comparing coefficients gives $A=-2, B=0, C=-24, D=3, E=-49$. So $y_p=-2x^4-24x^2+3x-49$.
Answer: (b) $-2x^4-24x^2+3x-49$

13. If z_1 and z_2 are real and equal roots, solution is
Solution: For repeated roots, solution is $y=(c_1+c_2x)e^{z_1x}$
Answer: (a) $y=(c_1+xc_2)e^{z_1x}$

14. Integrating factor for $(1+xy)y\,dx+(1-xy)x\,dy=0$
Solution: Check exactness: $M_y=1+2xy$, $N_x=1-2xy$. Not exact. Multiplying by

$\frac{1}{2x^2 y^2}$ makes it exact.

Answer: (b) $\frac{1}{2x^2 y^2}$

15. If $f(x)=|x|$ on $[0,2\pi)$, then a_0 is

    Solution: On $[0,2\pi)$, $|x|=x$. $a_0=\frac{1}{\pi}\int_0^{2\pi} x\,dx=\frac{1}{\pi}\cdot\frac{(2\pi)^2}{2}=2\pi$

    Answer: (b) 2π

16. If roots are 2,2,3, solution is
    Solution: For repeated root 2: $(A+Bx)e^{2x}$, plus Ce^{3x}
    Answer: (c) $y=(A+xB)e^{2x}+Ce^{3x}$

17. Fourier coefficient a_n for $f(x)=e^x$ half-range cosine on $(0,1)$

    Solution: $a_n=2\int_0^1 e^x \cos(n\pi x)\,dx=\frac{2}{1+n^2\pi^2}[(-1)^n e-1]$

    Answer: (a) $\frac{2}{1+n^2\pi^2}[(-1)^n e-1]$

18. Integrating factor for $(xy^2-e^{1/x^2})dx-x^2y\,dy=0$
    Solution: $M_y=2xy$, $N_x=-2xy$. Not exact. Try IF = x^{-4}: Multiply: $x^{-3}y^2dx-x^{-4}e^{1/x^2}dx-x^{-2}y\,dy$. Now $M_y=2x^{-3}y$, $N_x=2x^{-3}y$ (from $-x^{-2}y$ derivative). So exact.
    Answer: (c) x^{-4}

19. Orthogonal trajectories for $y^2=cx$

    Solution: Differentiate: $2yy'=c\Rightarrow y'=\frac{c}{2y}$. But $c=y^2/x$, so $y'=\frac{y}{2x}$. Orthogonal slope = $-\frac{2x}{y}$. Solve: $y\,dy=-2x\,dx\Rightarrow\frac{y^2}{2}=-x^2+C\Rightarrow x^2+\frac{y^2}{2}=C$

    Answer: (b) $x^2+\frac{y^2}{2}=c$

20. Solution of $\frac{dy}{dx}=\frac{x^2+y^2}{xy}$
    Solution: Homogeneous: let $y=vx$, $y'=v+xv'$. Then $v+xv'=\frac{1+v^2}{v}\Rightarrow xv'=\frac{1}{v}\Rightarrow v\,dv=\frac{dx}{x}$. Integrate: $\frac{v^2}{2}=\ln x+C$. But $v=y/x$, so $\frac{y^2}{2x^2}=\ln x+C$

    Answer: (b) $\log x+\frac{y^2}{2x^2}=c$

21. Complementary function for $y''+4y=0$
Solution: Aux eq: $m^2+4=0 \Rightarrow m=\pm 2i$. CF = $A\cos 2x+B\sin 2x$
Answer: (c) $y=A\cos 2x+B\sin 2x$

22. Particular integral for $(D^3-3D^2+4D-2)y=e^x+\cos x$
Solution: For e^x: $f(1)=0$, $f'(1)=1$, so PI = xe^x. For $\cos x$: Replace $D^2=-1$, get $1/(3D+1)\cos x=\frac{1}{10}(3\sin x+\cos x)$. Total = $xe^x+\frac{1}{10}(3\sin x+\cos x)$
Answer: (a) $xe^x+\frac{1}{10}(3\sin x+\cos x)$

23. Fourier sine transform of $f(x)=1/x$
Solution: $F_s(\omega)=\sqrt{\frac{2}{\pi}}\int_0^\infty \frac{\sin(\omega x)}{x}dx=\sqrt{\frac{2}{\pi}}\cdot\frac{\pi}{2}=\sqrt{\frac{\pi}{2}}$
Answer: (c) $\sqrt{\pi/2}$

24. Solution of Bernoulli equation $(xy^5+y)dx=dy$
Solution: Rewrite: $\frac{dy}{dx}-y=xy^5$. Divide by y^5: $y^{-5}y'-y^{-4}=x$. Let $v=y^{-4}$, then $v'=-4y^{-5}y'$, so $-\frac{v'}{4}-v=x \Rightarrow v'+4v=-4x$. Linear in v, IF = e^{4x}. Solve: $ve^{4x}=-xe^{4x}+\frac{e^{4x}}{4}+C \Rightarrow y^{-4}e^{4x}=-xe^{4x}+\frac{e^{4x}}{4}+C$
Answer: $y^{-4}e^{4x}=-xe^{4x}+\frac{e^{4x}}{4}+c$

25. Orthogonal trajectory for $e^x+e^{-y}=c$
Solution: Differentiate: $e^x dx-e^{-y}dy=0 \Rightarrow \frac{dy}{dx}=e^{x+y}$. Orthogonal slope = $-e^{-(x+y)}$.
Solve: $\frac{dy}{dx}=-e^{-x}e^{-y} \Rightarrow e^y dy=-e^{-x}dx \Rightarrow e^y=e^{-x}+C \Rightarrow e^y-e^{-x}=C$
Answer: (b) $e^x-e^{-y}=c$

26. Fourier coefficient a_n for $f(x)=x+x^2$ on $[-\pi,\pi)$
Solution: $a_n=\frac{1}{\pi}\int_{-\pi}^{\pi}(x+x^2)\cos(nx)dx$. $x\cos(nx)$ is odd ⇒ integral = 0. $x^2\cos(nx)$ is even, so $a_n=\frac{2}{\pi}\int_0^\pi x^2\cos(nx)dx=\frac{4(-1)^n}{n^2}$
Answer: (a) $\frac{4(-1)^n}{n^2}$

27. Given Fourier series for x^2, find $S=\frac{1}{1^2}-\frac{1}{2^2}+\frac{1}{3^2}-\frac{1}{4^2}+\cdots$

Solution: Put $x=0$ in given series: $0=\frac{\pi^2}{3}-4S \Rightarrow 4S=\frac{\pi^2}{3} \Rightarrow S=\frac{\pi^2}{12}$

Answer: (d) $\frac{\pi^2}{12}$

28. Complete solution for $p-\frac{1}{p}=\frac{x}{y}-\frac{y}{x}$

Solution: This equation factors into $(xy-c)(x^2-y^2-c)=0$
Answer: (b) $(xy-c)(x^2-y^2-c)$

29. General solution of $p=\cos(y-xp)$
Solution: Rewrite as $y=xp+\cos^{-1}p$. This is Clairaut's form, so replace p by c:
$y=cx+\cos^{-1}c$
Answer: (b) $y=cx+\cos^{-1}c$

30. Fourier coefficient b_n for $f(x)=x\sin x$ on $(-\pi,\pi)$
Solution: $f(-x)=(-x)\sin(-x)=(-x)(-\sin x)=x\sin x=f(x)$, so even function. For even function, $b_n=0$.
Answer: (i) 0

31. Order of $xy\frac{d^2y}{dx^2}+x\left(\frac{dy}{dx}\right)^2-y\frac{dy}{dx}=0$

Solution: Highest derivative is y'', so order = 2.
Answer: (ii) 2

32. $f(x,y)=3x^3+xy^2+\sin x+e^x$ is
Solution: Terms $\sin x$ and e^x are not homogeneous, so function is not homogeneous.
Answer: (ii) not a homogeneous function

33. Value of Wronskian $W(x,x^2,x^3)$
Solution:

$$W=\begin{vmatrix} x & x^2 & x^3 \\ 1 & 2x & 3x^2 \\ 0 & 2 & 6x \end{vmatrix}=x(12x^2-6x^2)-x^2(6x-0)+x^3(2-0)=6x^3-6x^3+2x^3=2x^3$$

Answer: (ii) $2x^3$

34. For $(2x+y)dx=(-x-y)dy$, check condition
Solution: $M=2x+y$, $N=x+y$. $\frac{\partial M}{\partial y}=1$, $\frac{\partial N}{\partial x}=1$. So $\frac{\partial M}{\partial y}=\frac{\partial N}{\partial x}$

Answer: (c) $\frac{\partial M}{\partial y}=\frac{\partial N}{\partial x}$

35. Bernoulli’s equation reduces to linear by substitution $u = y^{1-n}$ giving
$\frac{du}{dx} + (1-n)P(x)u = (1-n)Q(x)$
Answer: (b)

36. Solution of $\frac{dy}{dx} = -\frac{x}{y}$

Solution: Separate: $y\,dy = -x\,dx \Rightarrow \frac{y^2}{2} = -\frac{x^2}{2} + C \Rightarrow x^2 + y^2 = 2C = a^2$

Answer: (d) $x^2 + y^2 = a^2$

37. $x\,dy + y\,dx = d(xy)$
Answer: (a)

38. $\frac{y\,dx - x\,dy}{y^2} = d\left(\frac{x}{y}\right)$
Answer: (d)

39. $\frac{x\,dy - y\,dx}{x^2 + y^2} = d\left(\tan^{-1}\frac{y}{x}\right)$
Answer: (c)

40. Integrating factor for $f_1(xy)\,y\,dx + f_2(xy)\,x\,dy = 0$ is $\frac{1}{Mx - Ny}$

Answer: (a)

41. Orthogonal trajectories of family $F(x, y, c) = 0$ replace $\frac{dy}{dx}$ by $-\frac{dx}{dy}$

Answer: (d)

42. Two families cutting at right angles are orthogonal trajectories
Answer: (c)

43. $y\,dx + x\,dy = 0$ is exact
Solution: $M = y$, $N = x$, $M_y = 1$, $N_x = 1$ ⇒ exact
Answer: (a)

44. If $\frac{1}{M}\left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) = f(y)$, then IF = $e^{\int f(y)\,dy}$
Answer: (c)

45. Function F such that $FM\,dx + FN\,dy = 0$ is exact is integrating factor
Answer: (c)

46. Solution of $y\,dx + x\,dy = 0$ is $xy = c$
Answer: (a)

47. IF for homogeneous equation is $\frac{1}{Mx-Ny}$
Answer: (a)

48. Roots 2,2,3 ⇒ solution
Answer: (c) $y=(A+xB)e^{2x}+Ce^{3x}$

49. CF for $y''+4y=0$
Answer: (c) $y=A\cos 2x+B\sin 2x$

50. PI for $(D^3-3D^2+4D-2)y=e^x+\cos x$
Answer: (a) $xe^x+\frac{1}{10}(3\sin x+\cos x)$

51. Equation $xyp^2+p(3x^2-2y^2)-6xy=0$ is solvable for p
Answer: (a) p

52. Solution of $y=px+a$ is $y=cx+a$
Answer: (b)

53. Equation $3x^4p^2-xp-y=0$ is solvable for p
Answer: (a) p

54. Equation $p^2+xp+y=0$ is solvable for p
Answer: (a) p

55. Solution of $y=xp+p^2+2$ is $y=xc+c^2+2$
Answer: (c)

56. CF of $(D^2-5D+6)y=e^{4x}$ is $c_1e^{2x}+c_2e^{3x}$
Answer: (d)

57. CF is general solution of homogeneous linear DE
Answer: (a)

58. For real distinct roots: $c_1e^{m_1x}+c_2e^{m_2x}+\cdots$
Answer: (c)

59. Roots 2,2,3 ⇒ CF
Answer: (a) $(c_1+c_2x)e^{2x}+c_3e^{3x}$

60. Roots 2, -3, -3, -3 ⇒ CF
Answer: (a) $(c_1x^2+c_2x+c_3)e^{-3x}+c_4e^{2x}$

61. Roots $2\pm 3i$ and -3 ⇒ CF
Answer: (b) $e^{2x}(c_1\cos 3x+c_2\sin 3x)+c_3e^{-3x}$

62. Roots $a \pm i\beta$ repeated twice and -3 ⇒ CF
Answer: (a) $e^{ax}\left[\left(c_1+c_2x\right)\cos\beta x+\left(c_3+c_4x\right)\sin\beta x\right]+c_5e^{-3x}$

63. $y_p=\dfrac{1}{D-a}e^{ax}=x\,e^{ax}$
Answer: (b)

64. $y_p=\dfrac{1}{\left(D-a\right)^n}e^{ax}=\dfrac{x^n}{n!}e^{ax}$
Answer: (c)

65. $y_p=\dfrac{1}{D^2+a^2}\sin\left(ax+b\right)=-\dfrac{x\cos\left(ax+b\right)}{2a}$
Answer: (d)

66. $y_p=\dfrac{1}{\left(D^2+a^2\right)^n}\sin\left(ax\right)=\dfrac{x^n}{\left(2a\right)^n n!}\sin\left(ax-\dfrac{n\pi}{2}\right)$
Answer: (a)

67. $y_p=\dfrac{1}{\left(D^2+a^2\right)^n}\cos\left(ax+b\right)=\dfrac{x^n}{\left(2a\right)^n n!}\cos\left(ax-\dfrac{n\pi}{2}\right)$
Answer: (b)

68. $y_p=\dfrac{1}{f\left(D\right)}e^{ax}V\left(x\right)=e^{ax}\dfrac{1}{f\left(D+a\right)}V\left(x\right)$
Answer: (a)

69. $y_p=\dfrac{1}{f\left(D\right)}e^{ax+b}=\dfrac{e^{ax+b}}{f\left(a\right)}$
Answer: (a)

70. Equation $y=px+f\left(p\right)$ is Clairaut's equation
Answer: (b)

71. Complete solution of non-homogeneous DE = CF + PI
Answer: (d)

72. CF of $\left(D^2+1\right)y=\sin x$ is $a_1\cos x+a_2\sin x$
Answer: (a)

73. Solution with as many arbitrary constants as order is complete primitive
Answer: (b)

74. Solution of $\left(x^2-ay\right)dx=\left(ax-y^2\right)dy$
Solution: Exact: $M_y=-a$, $N_x=-a$. Integrate: $\int M\,dx=\dfrac{x^3}{3}-axy+\phi\left(y\right)$. Differentiate:

$-ax+\phi'(y)=-ax+y^2 \Rightarrow \phi'(y)=y^2 \Rightarrow \phi(y)=\frac{y^3}{3}+C$. So
$\frac{x^3}{3}-axy+\frac{y^3}{3}=C \Rightarrow x^3-3axy+y^3=3C=c$
Answer: (a) $x^3-3axy+y^3=c$

75. IF for $(y^2-xy)dx+x^2dy=0$
Solution: $M_y=2y-x$, $N_x=2x$. Not exact. Try IF = $1/(xy^2)$: Multiply: $(1/x-1/y)dx+(x/y^2)dy=0$. Now $M_y=1/y^2$, $N_x=1/y^2$ ⇒ exact.
Answer: (a) $\frac{1}{xy^2}$

76. IF for $(1+xy)ydx+(1-xy)xdy=0$
Answer: (b) $\frac{1}{2x^2y^2}$

77. IF for $(xy^2-e^{1/x^2})dx-x^2ydy=0$
Answer: (c) x^{-4}

78. Orthogonal trajectories for $y^2=cx$
Answer: (b) $x^2+\frac{y^2}{2}=c$

79. Solution of $\frac{dy}{dx}=\frac{x^2+y^2}{xy}$
Answer: (b) $\log x+\frac{y^2}{2x^2}=c$

80. DE of circles $x^2+y^2=a^2$
Solution: Differentiate: $2x+2yy'=0 \Rightarrow x+yy'=0 \Rightarrow yy'+x=0$
Answer: (c) $y\frac{dy}{dx}+x=0$

81. IF for $(xy^3+y)dx+2(x^2y^2+x+y^4)dy=0$
Answer: (c) $\frac{1}{y}$

82. Solution of $e^y dx+(2y+xe^y)dy=0$
Solution: Exact: $M_y=e^y$, $N_x=e^y$. Integrate $\int M\,dx=xe^y+\phi(y)$. Differentiate: $xe^y+\phi'(y)=2y+xe^y \Rightarrow \phi'(y)=2y \Rightarrow \phi(y)=y^2+C$. So $xe^y+y^2=C$
Answer: (a) $xe^y+y^2=c$

83. Solution of $(y-px)(p-1)=p$
Solution: Rewrite as $y=px+\frac{p}{p-1}$. Clairaut: replace p by c: $y=cx+\frac{c}{c-1}$. Multiply:

$y(c-1)=c(c-1)x+c \Rightarrow (c-1)y=c(c-1)x+c$. Rearranged: $(y-cx)(c-1)=c$
Answer: (a) $(y-cx)(c-1)=c$

84. PI of $(D^2+a^2)y=\sin ax$

Solution: When $D^2=-a^2$, formula: $-\frac{x}{2a}\cos ax$

Answer: (c) $-\frac{x}{2a}\cos ax$

85. PI of $\frac{d^2y}{dx^2}-2\frac{dy}{dx}+y=\cos 3x$

Solution: Replace $D^2=-9$: $\frac{1}{-9-2D+1}\cos 3x=\frac{1}{-2D-8}\cos 3x=-\frac{1}{2(D+4)}\cos 3x$.
Multiply numerator and denominator by $D-4$:
$-\frac{D-4}{2(D^2-16)}\cos 3x=-\frac{D-4}{2(-9-16)}\cos 3x=-\frac{D-4}{2(-25)}\cos 3x=\frac{D-4}{50}\cos 3x=\frac{1}{50}(-3\sin 3x-4\cos 3x)$
Answer: (c) $\frac{1}{50}(-3\sin 3x-4\cos 3x)$

86. $y_p=\frac{1}{f(D)}xV(x)=x\frac{1}{f(D)}V(x)-\frac{f'(D)}{[f(D)]^2}V(x)$
Answer: (d)

87. Solution of $\frac{dy}{dx}=\log\left(x\frac{dy}{dx}-y\right)$

Solution: Let $p=dy/dx$. Then $p=\log(xp-y) \Rightarrow xp-y=e^p \Rightarrow y=xp-e^p$. Clairaut: replace p by c: $y=cx-e^c$
Answer: (a) $y=cx-e^c$

88. CF for $\frac{d^2y}{dx^2}-8\frac{dy}{dx}+15y=0$

Solution: Aux eq: $m^2-8m+15=0 \Rightarrow m=3,5$. CF = $c_1e^{3x}+c_2e^{5x}$

89. PI for $\frac{d^2y}{dx^2}+16y=\sin 4x$

Solution: When $D^2=-16$, formula: $-\frac{x}{2\cdot 4}\cos 4x=-\frac{x}{8}\cos 4x$

90. Complete solution for $p-\frac{1}{p}=\frac{x}{y}-\frac{y}{x}$
Answer: (b) $(xy-c)(x^2-y^2-c)=0$

91. Solution of $\frac{d^2 y}{d x^2}-2\frac{d y}{d x}+5 y=0$

Solution: Aux eq: $m^2-2m+5=0 \Rightarrow m=1\pm 2i$. CF = $e^x\left(c_1\cos 2x+c_2\sin 2x\right)$
Answer: (b) $e^x\left(c_1\cos 2x+c_2\sin 2x\right)$

92. In Fourier series of $f(x)=\frac{1}{2}(\pi-x), 0<x<2\pi$, coefficient b_n

Solution: Known series: $\frac{\pi-x}{2}=\sum_{n=1}^{\infty}\frac{\sin n x}{n}$, so $b_n=\frac{1}{n}$

93. Given Fourier series leads to $\frac{\pi^2}{6}=1+\frac{1}{2^2}+\frac{1}{3^2}+\cdots$

Answer: (a)

94. Coefficient b_3 for given piecewise function

Solution: $b_3=\frac{1}{\pi}\left[\int_0^{\pi}(-1)\sin 3x\,dx+\int_{\pi}^{2\pi}2\sin 3x\,dx\right)=\frac{1}{\pi}\left[-\frac{2}{3}-\frac{4}{3}\right)=-\frac{2}{\pi}$

Answer: (a) $-\frac{2}{\pi}$

95. Coefficient a_2 for $f(x)=x^2, 0<x<4$

Solution: $a_2=\frac{2}{4}\int_0^4 x^2\cos\left(\frac{2\pi\cdot 2x}{4}\right)dx=\frac{1}{2}\int_0^4 x^2\cos(\pi x)\,dx=\frac{4}{\pi^2}$

96. If f is even, $\int_{-l}^{l} f(x)\,dx=2\int_0^{l} f(x)\,dx$

97. In Fourier cosine series of $f(x)=1, 0<x<\pi$, coefficient a_n

Solution: $a_n=\frac{2}{\pi}\int_0^{\pi}\cos(n x)\,dx$

Answer: (b)

98. Fourier coefficient b_n for $f(x)=x^3\sin x$ on $(-\pi,\pi)$
Solution: Even function $\Rightarrow b_n=0$
Answer: (a) 0

99. Even function in $[-\pi,\pi)$
Solution: $x^3\sin x$ is even because $(-x)^3\sin(-x)=-x^3(-\sin x)=x^3\sin x$
Answer: (b) $x^3\sin x$

100. In Fourier series of $f(x)=x, 0<x<2\pi$, coefficient of $\sin 2x$

Solution: $b_2=\frac{1}{\pi}\int_0^{2\pi}x\sin 2x\,dx=\frac{1}{\pi}[-\pi]=-1$

Answer: (d) -1

101. For $f(x)=x^2+x^4$ on $[-\pi,\pi)$, b_n is
Solution: Even function $\Rightarrow b_n=0$
Answer: (c) 0

102. Fourier transform of $f(3x)$
Solution: Scaling property: $F[f(ax)]=\frac{1}{|a|}F\left(\frac{s}{a}\right)$. Here $a=3$, so = $\frac{1}{3}F\left(\frac{s}{3}\right)$
Answer: (d) $\frac{1}{3}F\left(\frac{s}{3}\right)$

103. Given square wave series, sum $1-\frac{1}{3}+\frac{1}{5}-\frac{1}{7}+\cdots$
Solution: At $x=\pi/2$, $f(\pi/2)=\pi/4$. Series at $x=\pi/2$ gives $1-\frac{1}{3}+\frac{1}{5}-\frac{1}{7}+\cdots=\frac{\pi}{4}$
Answer: (b) $\frac{\pi}{4}$

104. Bernoulli equation $\frac{dy}{dx}+\frac{y}{x}=x^2y^2$ has coefficient in linearized form
Solution: Let $v=y^{-1}$, then $dv/dx-\frac{1}{x}v=-x^2$. Coefficient of v is $-\frac{1}{x}=-x^{-1}$
Answer: (a) $-x^{-1}$

105. Solution of $y\log y\,dx+(x-\log y)\,dy=0$
Solution: $\frac{dx}{dy}+\frac{1}{y\log y}x=\frac{1}{y}$. IF = $e^{\int\frac{1}{y\log y}dy}=\log y$. Multiply: $\frac{d}{dy}(x\log y)=\frac{\log y}{y}$.
Integrate: $x\log y=\frac{(\log y)^2}{2}+C$
Answer: (b) $x\log y=\frac{(\log y)^2}{2}+c$

106. Order of $\left(\frac{d^2y}{dx^2}+2x\frac{dy}{dx}\right)^{3/2}=2x$
Solution: Highest derivative is y'', so order = 2
Answer: (a) 2

107. IF for $(x+1)\frac{dy}{dx}-y=e^x(x+1)^2$
Solution: Divide by $x+1$: $\frac{dy}{dx}-\frac{1}{x+1}y=e^x(x+1)$. $P(x)=-\frac{1}{x+1}$, IF = $e^{\int-\frac{1}{x+1}dx}=e^{-\ln(x+1)}=\frac{1}{x+1}$
Answer: (c) $\frac{1}{x+1}$

108. IF for $(x \log x)\frac{dy}{dx}+y=(\log x)^2$

Solution: Divide by $x \log x$: $\frac{dy}{dx}+\frac{1}{x\log x}y=\frac{\log x}{x}$. $P(x)=\frac{1}{x\log x}$, IF = $e^{\int \frac{1}{x\log x}dx}=e^{\ln(\log x)}=\log x$

Answer: (a) $\log x$

109. Using Parseval: $\int_0^\infty \frac{dx}{(x^2+1)^2}$

Solution: Fourier cosine transform of e^{-x} is $\sqrt{\frac{2}{\pi}}\frac{1}{s^2+1}$. Parseval: $\int_0^\infty \frac{2}{\pi}\frac{1}{(s^2+1)^2}ds=\int_0^\infty e^{-2x}dx=\frac{1}{2}\Rightarrow\int_0^\infty \frac{1}{(s^2+1)^2}ds=\frac{\pi}{4}$

Answer: (c) $\frac{\pi}{4}$

110. Fourier sine transform of $f(x)=e^{-2x}$

Solution: $F_s(\omega)=\sqrt{\frac{2}{\pi}}\int_0^\infty e^{-2x}\sin(\omega x)dx=\sqrt{\frac{2}{\pi}}\frac{\omega}{\omega^2+4}$

Answer: (b) $\sqrt{\frac{2}{\pi}}\frac{\omega}{\omega^2+4}$

111. Fourier series of x^2 on $(-\pi,\pi)$

Solution: Known: $x^2=\frac{\pi^2}{3}+4\sum_{n=1}^{\infty}\frac{(-1)^n}{n^2}\cos nx$

Answer: (d)

112. Fourier series form on $(0,2\pi)$ is $\frac{a_0}{2}+\sum_{n=1}^{\infty}(a_n\cos nx+b_n\sin nx)$

Answer: (C)

113. $a_0=\frac{1}{\pi}\int_a^{a+2\pi} f(x)dx$

Answer: (A)

114. Even function ⇒ only cosine terms

Answer: (B)

115. Odd function ⇒ only sine terms

Answer: (C)

116. $a_n = \frac{1}{L}\int_{-L}^{L} f(x)\cos\left(\frac{n\pi x}{L}\right)dx$
Answer: (A)

117. Half-range cosine series for even functions
Answer: (B)

118. Half-range sine series by odd extension
Answer: (B)

119. $b_n = \frac{2}{L}\int_{0}^{L} f(x)\sin\left(\frac{n\pi x}{L}\right)dx$
Answer: (C)

120. Fourier transform definition: $\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} f(x)e^{-i\omega x}dx$
Answer: (B)

121. Fourier transform applies to non-periodic functions
Answer: (B)

122. $F\{f'(x)\} = i\omega F\{f(x)\}$
Answer: (D)

123. $F\{f(x-a)\} = e^{-i\omega a}F\{f(x)\}$
Answer: (A)

124. Convolution: $(f*g)(x) = \int_{-\infty}^{\infty} f(t)g(x-t)dt$
Answer: (D)

125. Fourier sine transform: $\sqrt{\frac{2}{\pi}}\int_{0}^{\infty} f(x)\sin(\omega x)dx$

126. Parseval's identity: $\int_{-\infty}^{\infty} |F(\omega)|^2 d\omega = \int_{-\infty}^{\infty} |f(x)|^2 dx$ (with appropriate constants)
Answer: (A)

127. Odd function ⇒ cosine coefficients are zero
Answer: (A)

128. Half-range cosine series on $(0, L)$
Answer: (B)

129. For $f(x)=1$ on $(-\pi, \pi)$, $a_0 = 2$
Answer: (C) 2

130. For $f(x)=1$ on $(-\pi,\pi)$, $a_n=0$ for $n\geq 1$
Answer: (D) 0

131. Sine transform for odd extension
Answer: (A)

132. Cosine transform for even extension

133. For $f(x)=c$ on $(-\pi,\pi)$, $b_n=0$
Answer: (C) 0

134. Fourier transform converts time domain to frequency domain
Answer: (A)

135. For $f(x)=x$ on $(-\pi,\pi)$, $a_0=0$
Answer: (D) 0

136. For $f(x)=x$ on $(-\pi,\pi)$, which coefficients are zero?
Solution: x is odd, so all a_n (including a_0) are zero.
Answer: (C) Both a_n and a_0

137. Fourier sine coefficient for $f(x)=x$ on $(0,\pi)$
Solution: $b_n=\frac{2}{\pi}\int_0^{\pi} x\sin(nx)\,dx=\frac{2}{\pi}\left[-\frac{\pi(-1)^n}{n}\right)=\frac{2(-1)^{n+1}}{n}$
Answer: (D) $\frac{2(-1)^{(n+1)}}{n}$

138. For $f(x)=\sin x$ on $(-\pi,\pi)$, b_n is
Solution: Fourier series of $\sin x$ is itself, so $b_1=1$, others 0.
Answer: (A) 1 for $n=1$ only

139. For $f(x)=\cos 3x$ on $(-\pi,\pi)$, $a_3=1$

140. For $f(x)=x^3$ on $(-2,2)$, Fourier series contains
Solution: x^3 is odd, so only sine terms
Answer: (B) Sine terms

141. $F\{f(x-a)\}=e^{-i\omega a}F\{f(x)\}$
Answer: (C)

142. Order and degree of $\frac{d^2y}{dx^2}+\left(\frac{dy}{dx}\right)^2+y=0$
Solution: Order = 2, degree = 1 (all derivatives in polynomial form with power 1 for highest derivative)
Answer: (B) Order 2, degree 1

143. Homogeneous DE: $\frac{dy}{dx}=\frac{x^2+y^2}{xy}$

Solution: RHS is homogeneous of degree 0

Answer: (A)

144. Solution of $\frac{dy}{dx}=\frac{y}{x}$ is $y=Cx$

Answer: (D)

145. Bernoulli equation $\frac{dy}{dx}=3xy+4y^4$ has $n=4$

Answer: (A) 4

146. If $F(f(x))=\frac{1}{1+\omega^2}$, then $F(f')=i\omega F(f)=\frac{i\omega}{1+\omega^2}$

Answer: (A)

147. IF for $(1+x^2)\frac{dy}{dx}-2xy=2x$

Solution: Divide by $1+x^2$: $\frac{dy}{dx}-\frac{2x}{1+x^2}y=\frac{2x}{1+x^2}$. $P(x)=-\frac{2x}{1+x^2}$, IF = $e^{\int -\frac{2x}{1+x^2}dx}=e^{-\ln(1+x^2)}=\frac{1}{1+x^2}$

Answer: (A)

148. Solution of $\frac{dy}{dx}+y=e^x$

Solution: IF = e^x. Multiply: $\frac{d}{dx}(e^x y)=e^{2x}\Rightarrow e^x y=\frac{e^{2x}}{2}+C\Rightarrow y=\frac{e^x}{2}+Ce^{-x}$

Answer: (B)

149. For $\frac{dy}{dx}=y+xy^2$, let $v=1/y$, then $v'+v=-x$

150. Bernoulli equation form: $\frac{dy}{dx}+P(x)y=Q(x)y^n$

Answer: (B)

151. IF for $\frac{dy}{dx}+\frac{2}{x}y=x^2$ is x^2

152. For $f(x)=e^{-x}, 0\le x\le 2\pi$, $a_0=\frac{1}{\pi}(1-e^{-2\pi})$

Answer: (A)

153. For $f(x)=x\sin x, 0\le x\le 2\pi$, $a_0=-2$

Answer: (C) -2

154. For given piecewise function, $a_1=\frac{1}{2}$

Answer: (A)

155. Solution of $y=(x-a)p-p^2$
Answer: (c) $y=(x-a)c-c^2$

156. $(y+x)dx-(y-x)dy=0$ is both exact and homogeneous
Answer: (c)

157. IF for $(2xy)dx+(y^2-3x^2)dy=0$
Solution: $M_y=2x$, $N_x=-6x$. $(N_x-M_y)/M=(-8x)/(2xy)=-4/y$ (function of y alone). IF = $e^{\int -4/y dy}=e^{-4\ln y}=y^{-4}=\frac{1}{y^4}$

Answer: (b)

158. Solution of $p^2-7p+12=0$
Solution: $p=3$ or 4. Then $y=3x+c$ and $y=4x+c$. Combined: $(y-3x-c)(y-4x-c)=0$
Answer: (d) $(y-4x-c)(y-3x-c)=0$

159. IF for $(x^2+y^2+x)dx+xydy=0$
Solution: $M_y=2y$, $N_x=y$. $(M_y-N_x)/N=(2y-y)/(xy)=1/x$ (function of x alone). IF = $e^{\int 1/x dx}=e^{\ln x}=x$
Answer: (a) x

160. Clairaut's equation: $y=px+\log p$
Answer: (d)

161. Orthogonal trajectory of $\frac{dy}{dx}=x$ is $\frac{dy}{dx}=-\frac{1}{x}$

Answer: (c)

162. Equation $\frac{dy}{dx}+Py=Qy^n$ is Bernoulli's equation
Answer: (a)

163. IF for $\frac{dy}{dx}+Py=Q$ is $e^{\int P dx}$
Answer: (b)/(c)

164. IF for $\frac{dy}{dx}+\frac{y}{x}=x^3$ is x
Answer: (a)

165. Exact DE: $(2xy+y^2)dx+(x^2+2xy)dy=0$
Solution: $M_y=2x+2y$, $N_x=2x+2y$ ⇒ exact
Answer: (a)

166. IF for Bernoulli $x\frac{dy}{dx}+y=x^3y^6$
Solution: Divide by x: $\frac{dy}{dx}+\frac{1}{x}y=x^2y^6$. $n=6$, let $v=y^{-5}$. Then $v'-\frac{5}{x}v=-5x^2$. IF = $e^{\int -\frac{5}{x}dx}=e^{-5\ln x}=x^{-5}$
Answer: (a) x^{-5}

167. Solution of $\frac{dy}{dx}+\frac{y}{x}=x^2$
Solution: IF = x. Multiply: $\frac{d}{dx}(xy)=x^3 \Rightarrow xy=\frac{x^4}{4}+C$
Answer: (a)

168. Solution of $y'-y=e^{2x}$
Solution: IF = e^{-x}. Multiply: $\frac{d}{dx}(e^{-x}y)=e^x \Rightarrow e^{-x}y=e^x+C \Rightarrow y=e^{2x}+Ce^x$
Answer: (a)

169. Solution of $(x^2-y)dx+(y^2-x)dy=0$
Solution: Exact: $M_y=-1$, $N_x=-1$. Integrate: $\frac{x^3}{3}-xy+\frac{y^3}{3}=C$
Answer: (c)

170. Solution of $y''-5y'+6y=0$
Solution: Aux eq: $m^2-5m+6=0 \Rightarrow m=2,3$. CF = $c_1e^{2x}+c_2e^{3x}$
Answer: (b)

171. Solution of $y''-3y'-4y=0$
Solution: Aux eq: $m^2-3m-4=0 \Rightarrow m=4,-1$. CF = $c_1e^{4x}+c_2e^{-x}$
Answer: (a)

172. Solution of Clairaut $y=px+\frac{a}{p}$
Solution: Replace p by c: $y=cx+\frac{a}{c}$
Answer: (a)

173. Orthogonal trajectory of circles $x^2+y^2=a^2$
Solution: Differentiate: $2x+2yy'=0 \Rightarrow y'=-x/y$. Orthogonal slope = y/x. Solve: $dy/dx=y/x \Rightarrow \ln y=\ln x+C \Rightarrow y=Cx$ (straight lines)
Answer: (b)

174. For equal roots r, CF = $\left(C_1+C_2x\right)e^{rx}$
Answer: (b)

175. In variation of parameters, $u_1=-\int\frac{y_2X}{W}dx$
Answer: (a)

176. IF for homogeneous equation is $\frac{1}{Mx+Ny}$
Answer: (a)

177. $\frac{y\,dx-x\,dy}{y^2}=d\left(\frac{x}{y}\right)$
Answer: (d)

178. Solution of $y\,dx+x\,dy=0$ is $xy=c$
Answer: (a)

179. PI of $y''-4y'+4y=e^{-7t}$
Solution: Try Ae^{-7t}: $\left(49+28+4\right)Ae^{-7t}=81Ae^{-7t}=e^{-7t}\Rightarrow A=1/81$
Answer: (d) $\frac{e^{-7t}}{81}$

180. Solutions are LI if $W\neq0$
Answer: (c)

181. PI of $\left(D^2+4\right)y=\sin 2x$
Solution: When $D^2=-4$, formula: $-\frac{x}{2\cdot2}\cos 2x=-\frac{x}{4}\cos 2x$
Answer: (b)

182. For triple root 3, CF = $\left(C_1+C_2x+C_3x^2\right)e^{3x}$
Answer: (c)

183. Solution of $y''-2y'+2y=0$
Solution: Aux eq: $m^2-2m+2=0\Rightarrow m=1\pm i$. CF = $e^x\left(C_1\cos x+C_2\sin x\right)$
Answer: (c)

184. Solution of $\left(D^2-6D+8\right)y=e^{6x}$
Solution: CF = $c_1e^{2x}+c_2e^{4x}$. $f\left(6\right)=36-36+8=8\neq0$, so PI = $\frac{e^{6x}}{8}$
Answer: (b)

185. For $y''+y=sec\,x$, Wronskian = 1
Answer: (b) 1

186. PI of $(D^2+9)y=\sin 2x$

Solution: $f(-4)=-4+9=5\neq 0$, so PI = $\frac{\sin 2x}{5}$

Answer: (b) $-\frac{1}{5}\sin 2x$ (sign may be misprint)

187. For $y''-4y'=6x^2$, assume $y_p=Ax^3+Bx^2+Cx$

188. Wronskian of 1 and e^x is e^x
Answer: (a)

189. Best method for $y''+y=\tan x$ is variation of parameters
Answer: (b)

190. Solution of $(D^2+9)y=0$: $C_1\cos 3x+C_2\sin 3x$
Answer: (a)

191. If $f(a)=0$, PI involves factor x
Answer: (a)

192. For $(D-2)^2 y=17e^{2x}$, assume Ax^2e^{2x}
Answer: (c)

193. Roots of $D^2y+Dy+y=0$ are complex conjugates
Answer: (c)

194. Solution of $(x+y-10)dx+(x-y-2)dy=0$

Solution: Exact: $M_y=1$, $N_x=1$. Integrate: $\frac{x^2}{2}+xy-10x-\frac{y^2}{2}-2y=C$. Multiply by 2:

$x^2+2xy-20x-y^2-4y=2C$
Answer: (a)

195. Solution of Clairaut $p=\log(px-y)$
Solution: $px-y=e^p \Rightarrow y=px-e^p$. Replace p by c: $y=cx-e^c$
Answer: (b)

196. Orthogonal trajectories of $y=ax^2$
Solution: $y'=2ax=2y/x$. Orthogonal slope = $-x/(2y)$. Solve:

$2y\,dy=-x\,dx \Rightarrow y^2=-\frac{x^2}{2}+C \Rightarrow \frac{x^2}{2}+y^2=C$

Answer: (a) $\frac{x^2}{2}+y^2=b$

197. Solution of $(x^2-ay)dx=(ax-y^2)dy$
Answer: (a) $x^3-3axy+y^3=C$

198. IF for $(x+2)\frac{dy}{dx}-y=e^{x}(x+2)^{2}$

Answer: (b) $\frac{1}{x+2}$

199. Solution of $(2xy+y^{2})dx+(x^{2}+2xy)dy=0$

Answer: (c) $x^{2}y+xy^{2}=c$

200. IF for $y(y^{2}-2x^{2})dx+x(2y^{2}-x^{2})dy=0$

Answer: (c) $\frac{1}{2xy(y^{2}-x^{2})}$

201. Orthogonal trajectories for $x^{2}+4y^{2}=c$

Solution: $2x+8yy'=0\Rightarrow y'=-\frac{x}{4y}$. Orthogonal slope = $\frac{4y}{x}$. Solve:

$\frac{dy}{dx}=\frac{4y}{x}\Rightarrow\frac{dy}{y}=\frac{4dx}{x}\Rightarrow\ln y=4\ln x+C\Rightarrow y=Cx^{4}$

Answer: (a) $y=x^{4}+c$

202. Solution of $(D^{2}-3D+2)y=e^{3x}$

Answer: (b) $y=c_{1}e^{x}+c_{2}e^{2x}+\frac{e^{3x}}{2}$

203. PI of $(D^{2}+4)y=\sin 3x$

Answer: (a) $-\frac{1}{5}\sin 3x$

204. For $y''+y=sec\,x,\,u=\ln|\cos x)$

Answer: (c)

205. Solution of $(D^{2}+4)y=16$

Answer: (a) $(C_{1}\cos 2x+C_{2}\sin 2x)+4$

206. PI of $\frac{d^{2}y}{dx^{2}}-6\frac{dy}{dx}+9y=e^{3x}$

Answer: (b) $\frac{x^{2}e^{3x}}{2}$

207. Solution of $\frac{d^{2}y}{dx^{2}}+6\frac{dy}{dx}+9y=5e^{3x}$

Answer: (c) $(c_{1}+c_{2}x)e^{-3x}+\frac{5e^{3x}}{36}$

208. PI of $\frac{d^2 y}{d x^2}+4 y=e^{-2x}$

Answer: (b) $\frac{e^{-2x}}{8}$

209. PI of $\left(D^2-1\right) y=x$

Answer: (b) $-x$

210. Solution of $x\,d y-y\,d x+2 x^3 d x=0$

Solution: Divide by x^2:

$\frac{x\,d y-y\,d x}{x^2}=-2 x\,d x \Rightarrow d\left(\frac{y}{x}\right)=-2 x\,d x \Rightarrow \frac{y}{x}=-x^2+C \Rightarrow \frac{y}{x}+x^2=C$

Answer: (a) $\frac{y}{x}+x^2=c$

211. Solution of $\frac{d^2 y}{d x^2}+8 \frac{d y}{d x}+16 y=0$

Answer: (c) $\left(c_1+c_2 x\right) e^{-4x}$

212. Solution of $\left(D^2+9\right) y=\sin 4x$

Answer: (b) $c_1 \cos 3x+c_2 \sin 3x-\frac{1}{7} \sin 4x$

213. Orthogonal trajectory of $e^x+e^{-y}=c$

Answer: (b) $e^x-e^{-y}=c$